

Movie Rating Prediction using Machine Learning

1. Introduction

Movie recommendation systems are widely used by online streaming platforms to suggest movies to users based on their preferences. These systems analyze user behavior and previous ratings to predict how a user might rate a movie they have not seen yet.

In this project, a movie rating dataset was used to build a predictive model that estimates the rating a user may give to a movie. Machine learning techniques were applied to analyze the relationship between users and movies and generate predicted ratings.

2. Dataset Description

The dataset used for this analysis contains information about movie ratings given by users. It includes the following main components:

- **User ID** – Unique identifier for each user
- **Movie ID** – Unique identifier for each movie
- **Rating** – The score given by the user to the movie (usually between 0.5 and 5)
- **Timestamp** – The time when the rating was given

Additionally, the dataset may include a **movies file** containing movie titles and genres.

This dataset helps analyze user preferences and movie popularity.

3. Data Preprocessing

Before building the prediction model, several preprocessing steps were performed:

1. **Loading the dataset** into the Python environment using Pandas.
2. **Merging movie and rating datasets** to combine movie information with user ratings.

3. **Handling missing values** and removing unnecessary columns.
4. Creating a **user–movie rating matrix**, where rows represent users and columns represent movies.
5. Normalizing or organizing the data so that it can be used in a machine learning model.

These steps ensure that the data is clean and suitable for predictive analysis.

4. Model Implementation

To predict movie ratings, a **collaborative filtering approach** was applied. Collaborative filtering works by identifying similarities between users or items.

The model analyzes:

- Users who have similar movie preferences
- Movies that receive similar ratings from different users

Based on these similarities, the model predicts how a user might rate a movie they have not watched yet.

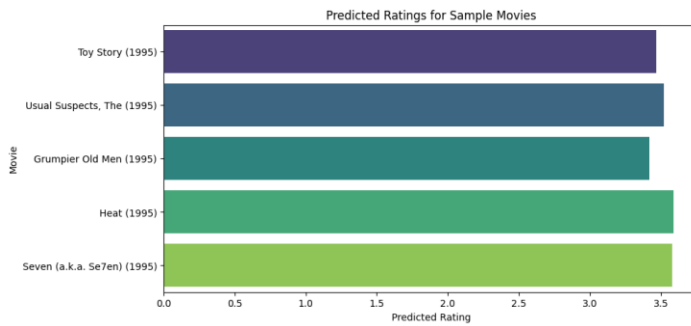
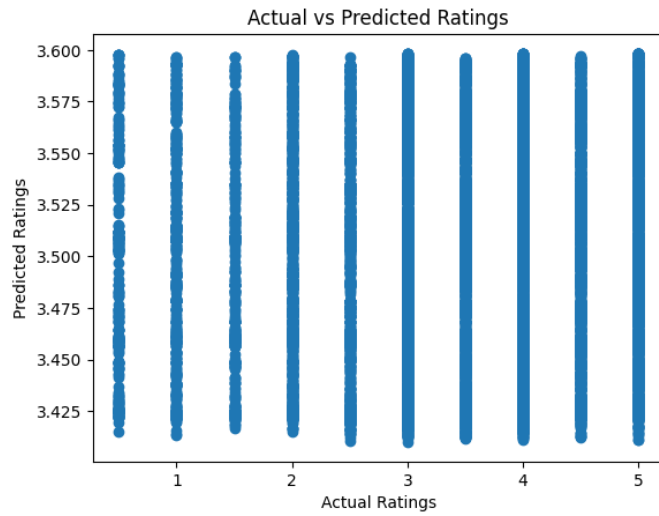
In some cases, regression models can also be used to estimate ratings by learning patterns from historical data.

5. Results and Visualization

After training the model, predicted ratings were generated for movies that users had not rated before.

The results were visualized using graphs such as:

- **Bar plots** showing predicted ratings for different movies
- **Rating distribution plots** to understand how users rate movies



These visualizations help understand which movies are likely to receive higher ratings from users.

The model performance was evaluated by comparing predicted ratings with actual ratings using evaluation metrics such as:

- **Mean Squared Error (MSE)**
- **Root Mean Squared Error (RMSE)**

Lower values of these metrics indicate better prediction accuracy.

6. Conclusion

In this project, a movie rating prediction model was developed using a movie ratings dataset. Data preprocessing, model building, and evaluation techniques were applied to estimate user preferences for unseen movies.

The results demonstrate that machine learning techniques can effectively predict user ratings and support recommendation systems. Such systems are widely used in platforms like streaming services to improve user experience and provide personalized recommendations.